Deep Learning-based Stock Market Prediction Using Historical Price and Topic-Distributed News Sentiment

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1. Introduction

Email remains one of the most essential and widely used modes of digital communication in both personal and professional contexts. Despite the rapid growth of instant messaging platforms, email continues to serve as a primary medium for official correspondence, academic communication, and business transactions. However, the increasing reliance on email has also led to a significant rise in unsolicited spam messages. These spam emails often contain phishing links, fraudulent offers, malicious attachments, or misleading promotional content, which can result in financial losses, data breaches, and reduced user productivity [1].

The dynamic and evolving nature of spam content poses a major challenge for automated spam filtering systems. Spam messages are intentionally designed to bypass detection mechanisms by exploiting variations in language, formatting, and semantic ambiguity. As a result, traditional rule-based or keyword-matching spam filters have become increasingly ineffective, as they rely heavily on manually crafted rules that lack adaptability to new spam patterns [2,3]. This limitation has motivated the adoption of data-driven approaches that can automatically learn discriminative features from text data.

In recent years, text classification techniques based on Natural Language Processing (NLP) have been widely applied to spam detection tasks. One common approach is the use of statistical text representations such as Term Frequency–Inverse Document Frequency (TF-IDF), which transforms textual data into numerical vectors by capturing the importance of words within a document relative to the entire corpus. TF-IDF remains a popular choice due to its simplicity, interpretability, and computational efficiency, particularly in environments with limited resources. However, TF-IDF is inherently a bag-of-words representation, which ignores word order and contextual relationships, making it less effective in capturing semantic meaning and linguistic nuances, especially in complex or informal text [2,4].

To overcome these limitations, Deep Learning (DL) models have gained increasing attention for text classification tasks. Convolutional Neural Networks (CNNs), originally developed for computer vision, have demonstrated strong performance in NLP applications by effectively capturing local n-gram features and hierarchical patterns within text sequences [5,6]. When combined with TF-IDF features, CNNs can serve as a powerful baseline model that balances efficiency and performance by learning discriminative patterns from sparse textual representations.

More recently, Transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) have achieved state-of-the-art performance across a wide range of NLP tasks. BERT leverages self-attention mechanisms to model contextual relationships between words, enabling a deeper understanding of semantics and syntax [7]. For Indonesian-language text, IndoBERT has demonstrated strong capability in modeling contextual and morphological characteristics of the language, which is particularly important for spam detection where informal expressions and abbreviations are frequently used [8].

Integrating contextual embeddings from IndoBERT with CNN architectures offers a promising hybrid approach. While IndoBERT provides rich, context-aware word representations, CNNs further enhance the model by extracting salient local features that are crucial for classification tasks. This combination allows the model to leverage both global contextual understanding and local pattern detection, resulting in improved robustness and generalization performance on unseen data [6,8].

Despite the growing adoption of deep learning techniques, research on Indonesian email spam classification remains relatively limited compared to high-resource languages such as English. Moreover, systematic comparisons between lightweight feature-based deep learning models, such as TF-IDF combined with CNN, and transformer-based hybrid architectures, such as IndoBERT combined with CNN, are still scarce. This gap highlights the need for empirical studies that analyze the trade-offs between computational efficiency, model complexity, and classification performance in the context of Indonesian-language spam detection [8,9].

Therefore, this study aims to evaluate and compare the performance of two deep learning pipelines for Indonesian email spam classification: TF-IDF combined with Convolutional Neural Networks and IndoBERT combined with Convolutional Neural Networks. By conducting experiments on an Indonesian spam email dataset [9], this research provides insights into how different text representation strategies influence model performance and highlights the effectiveness of contextual embeddings for spam detection in low-resource languages.

2. Dataset

The dataset used in this study was obtained from the "Indonesian Email Spam" collection available on the Kaggle platform. This dataset is a localized adaptation of an existing English-language spam dataset, which was translated into Indonesian to better reflect the linguistic characteristics and spam patterns commonly found in Indonesian email communication. The translation process preserves the semantic intent of each message, allowing the dataset to maintain the contextual nuances required for effective spam classification.

The dataset consists of labeled email text entries categorized into “spam” and “non-spam” classes. Each entry contains raw email content that varies in structure, tone, and length—ranging from personal correspondence to promotional messages and phishing attempts. This variability makes the dataset suitable for evaluating both traditional text-representation methods (such as TF-IDF) and modern deep learning approaches (such as BERT embeddings).

Before model development, the dataset underwent several preprocessing steps, including text normalization, noise removal (such as URLs, emojis, punctuation, and numerical artifacts), and tokenization. These steps were necessary to ensure consistency across samples and to enhance model learning efficiency.

Overall, the dataset provides a balanced and meaningful representation of real-world Indonesian email usage, making it an appropriate foundation for assessing the performance of different natural language processing (NLP) techniques in spam detection.

2.1. Exploratory Data Analysis

After the raw dataset is preprocessed, such as case folding, cleaned punctuation and symbols, removed stopwords, and finally stemmed the dataset. The final texts that preprocessed are compared with the raw texts in Table 1

PROCESSED TEXT

Raw Text	Preprocessed Text
Pemasangan peralatan yang Anda pesan selesai - - - Sistem Pemberitahuan Otomatis (Permintaan #: ECTH - 4 RSTT 6) Diminta untuk: Vince J Kaminski diminta oleh: Shirley Crenshaw Pemasangan peralatan (lihat di bawah) telah selesai. EN 6600 128 MB Dipasang di Desktop EN 6600 Desktop P 3 - 600 10 GB 64 MB 32 X NT 4. 0 EN 6600 128 MB Dipasang di Desktop	pemasangan peralatan pesan selesai sistem pemberitahuan otomatis permintaan ecth rstt diminta vince j kaminski diminta shirley crenshaw pemasangan peralatan lihat bawah selesai en mb dipasang desktop en desktop p gb mb x nt en mb dipasang desktop
Pemberitahuan Kegagalan Hai. Ini adalah program Qmail - Kirim di Bouncehost. Saya takut saya tidak dapat mengirimkan pesan Anda ke alamat berikut. Ini adalah kesalahan permanen ; aku sudah menyerah . Maaf itu tidak berhasil. : Maaf, tidak ada kotak surat di sini dengan nama itu. VPOPMAIL (# 5. 1. 1) - - - Terlampir adalah header asli dari pesan tersebut.	pemberitahuan kegagalan hai program qmail kirim bouncehost takut mengirimkan pesan alamat berikut kesalahan permanen aku menyerah maaf berhasil maaf kotak surat sini nama vpopmail terlampir header asli pesan tersebut

Next, the dataset was analyzed, and insight about the dataset label is shown below.

- Ham Label

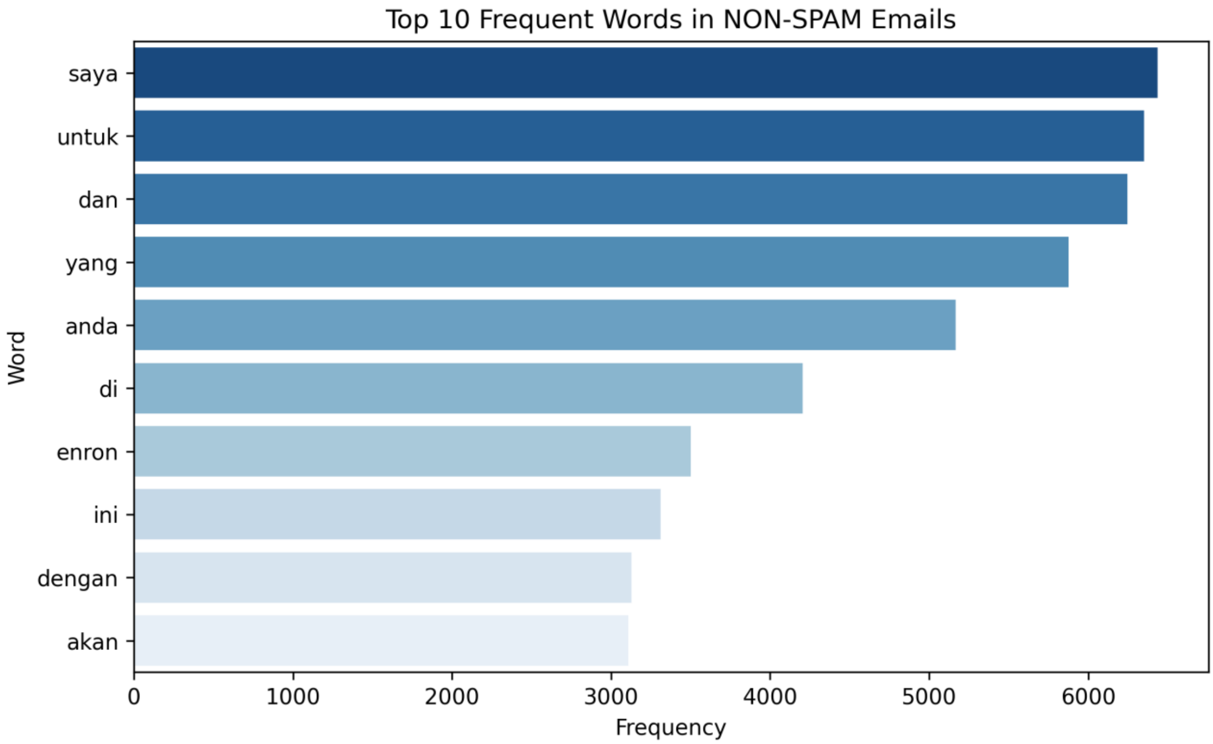

Fig. 3. Top words for ham label

The bar chart for the ham label shows the most frequent words appearing in non-spam emails. Common terms such as “saya”, “untuk”, and “dan” reflect natural conversational or formal writing patterns typically found in regular communication. These frequent words suggest that ham messages mostly consist of everyday exchanges, informational notices, or general conversation rather than promotional or suspicious content.

- Spam Label

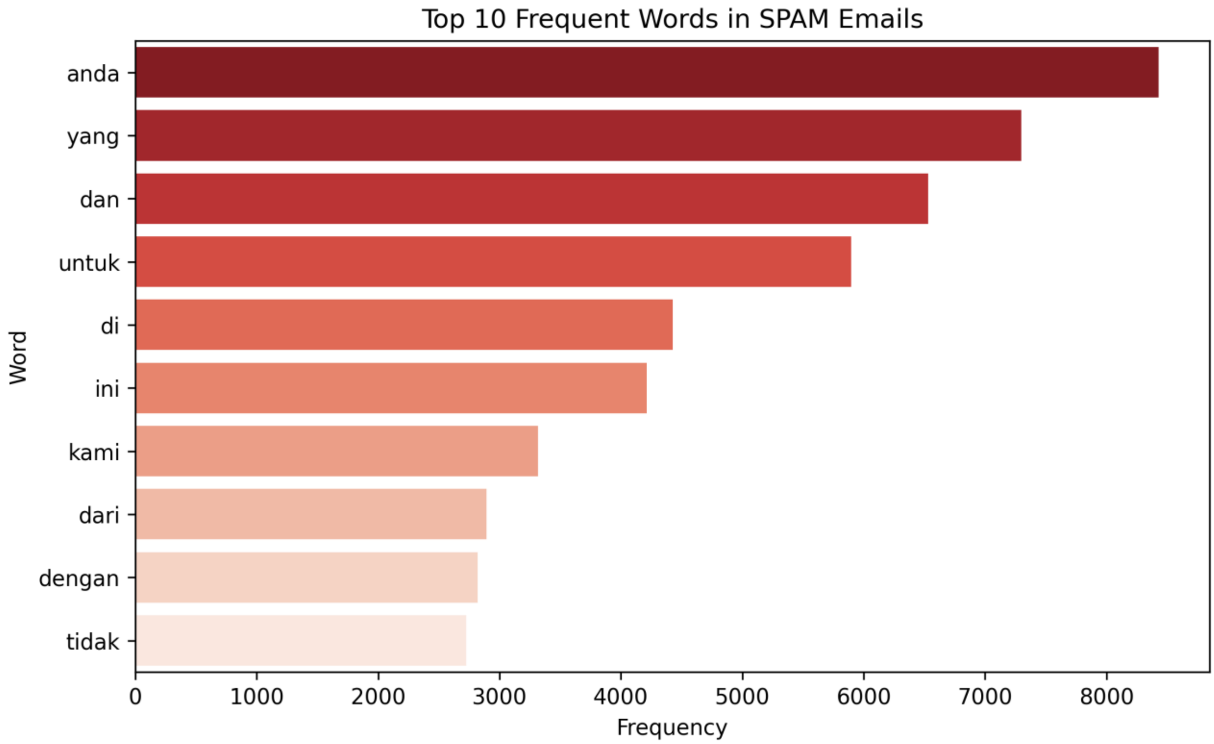

Fig. 5. Top words for spam label

The bar chart for the spam label shows common words frequently appearing in spam emails, such as “anda”, “yang”, and “dan”. These terms suggest that spam messages often use general, broad language aimed at addressing recipients directly and creating persuasive or attention-grabbing statements. The appearance of words like “untuk”, “kami”, and “dari” indicates message structures commonly used in promotional, persuasive, or unsolicited offers. Overall, the frequent words show that spam emails tend to adopt formal, yet generic language patterns designed to sound convincing to a wide audience.

3. Modelling

This research will go through five stages, starting with data collection, followed by data preprocessing which involves tasks such as sentence extraction, lowercasing and punctuation removal and stemming to identify meaningful words. Then followed by exploratory data analysis. After that, applying feature extraction the data is split into training and testing sets for model development. The modelling evaluation to use accuracy, precision, recall, and F1-score. The final stage results interpretation to identify if the email is a spam email or not. Fig.1 is the research workflow that is used in this study:

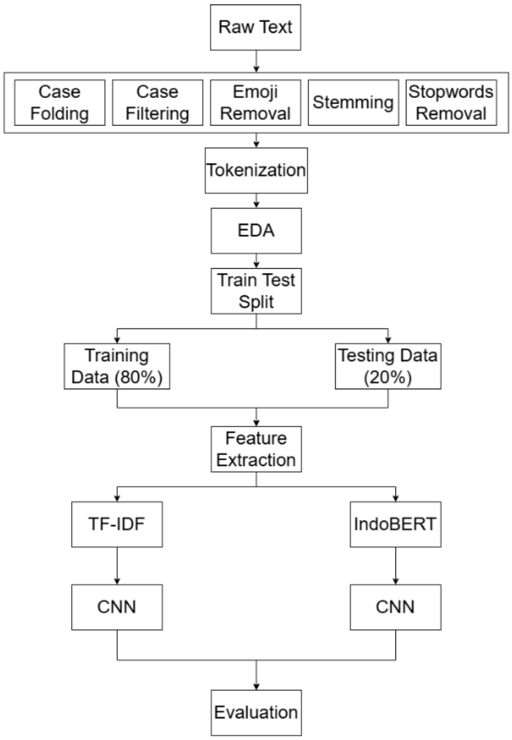

Fig. 1. General architecture.

3.1. Data Preprocessing

For this dataset, data used for the need to be simplified before research. In the preprocessing phase, tasks case folding, case filtering, emoji removal, stemming and stopwords removal. All these steps are to elevate the quality of this dataset for optimal performance.

3.2. Text Tokenization

After completing the text preprocessing stage, the next essential step in preparing the email data for classification is text tokenization. Tokenization refers to the process of breaking down a sequence of text into smaller, meaningful units known as tokens. These tokens can represent words, subwords, or even characters, depending on the chosen tokenization strategy.

In this study, tokenization serves a critical role because machine learning and deep learning models cannot directly interpret raw text data. Instead, the models operate on numerical representations derived from these tokens. By converting the text into a structured sequence of tokens, the models can effectively process linguistic patterns such as word frequency, position, and context.

For the TF-IDF + CNN model, tokenization mainly involves splitting sentences into individual words using a standard tokenizer. Each distinct word is mapped to its corresponding index within the vocabulary. These tokens are then transformed into TF-IDF feature vectors, capturing the importance of each word relative to the email corpus.

In contrast, for the IndoBERT + CNN architecture, tokenization is performed using the IndoBERT tokenizer, which employs a WordPiece subword algorithm. This tokenizer breaks words into smaller subword units to better handle rare or out-of-vocabulary words, improving the model's ability to generalize across diverse email content. The output includes token IDs, attention masks, and segment embeddings that align with the BERT input format.

Through tokenization, the email text is converted into a standardized numerical form while preserving essential semantic and syntactic information. This step is fundamental to ensuring that both traditional and transformer-based models can effectively interpret and learn from the underlying patterns within the dataset.

3.3. Data Splitting

After completing the preprocessing stage, the dataset must be divided into separate subsets to ensure that the model is trained and evaluated properly. In this study, the dataset is split into three parts: a training set, a validation set, and a testing set, following an 80:10:10 ratio.

The training set (80%) represents the majority of the data and is used to fit the model parameters. During this phase, the model learns patterns, word distributions, and relationships within the email texts that help distinguish between spam and non-spam messages.

The validation set (10%) is used during the training process but remains separate from the training data itself. Its purpose is to monitor the model's performance on unseen data, enabling hyperparameter tuning, and preventing overfitting. By evaluating the model on the validation set after each epoch, we can determine whether the model is learning effectively or if adjustments are needed to improve generalization.

Finally, the testing set (10%) remains completely unseen during training and validation. Once model training is complete, this set is used to provide an unbiased assessment of the model's real-world performance. Evaluating the model on this withheld data ensures that the reported results reflect its actual ability to classify new, previously unseen emails.

This 80–10–10 splitting strategy is widely used in machine learning because it provides sufficient data for training while preserving reliable subsets for validation and final evaluation. Through this structured data partitioning, the study ensures a fair and accurate comparison between the TF-IDF + CNN model and the embedding-based approach.

3.4. Feature Extraction

Feature extraction plays a critical role in transforming raw textual email data into numerical representations that can be processed by deep learning models. In this project, two different feature extraction approaches were implemented, each aligned with a distinct model pipeline used to evaluate spam classification performance. These approaches are TF-IDF combined with a Convolutional Neural Network (CNN), and IndoBERT embeddings combined with CNN. Together, they enable a comparative study between traditional keyword-driven representations and modern contextualized language embeddings within the Indonesian spam email domain.

The first feature extraction method employed is TF-IDF (Term Frequency–Inverse Document Frequency), a classical statistical technique widely used in information retrieval and text mining. TF-IDF represents each document as a high-dimensional vector, where each dimension corresponds to a word's importance relative to the entire corpus. After performing text cleaning, including case normalization, removing links, emojis, punctuation, and stopwords, the TF-IDF vectorizer encodes the processed text into a sparse numerical format. Unlike traditional machine learning approaches that typically combine TF-IDF with algorithms such as Naïve Bayes or SVM, this project integrates TF-IDF directly into a deep learning architecture. Specifically, the sparse vectors are reshaped into a format compatible with convolutional layers and fed into a CNN. The CNN then learns discriminative local patterns based on TF-IDF activations, effectively capturing keyword-based signals indicative of spam behavior. This hybrid approach leverages both the interpretability of TF-IDF and the pattern-learning ability of CNNs.

The second feature extraction approach utilizes IndoBERT, a transformer-based language model pretrained on large Indonesian textual corpora. IndoBERT captures far richer linguistic information by producing contextual embeddings that reflect not only the identity of words but also their semantic relationships within sentences. Instead of relying on statistical frequency counts like TF-IDF, IndoBERT generates dense representations for each token that encode grammar, meaning, and sentence structure. After preprocessing and tokenization using the IndoBERT tokenizer, each email is transformed into sequences of contextual embeddings. These embeddings are subsequently passed into a CNN layer, which extracts task-relevant features from the transformer output. This architecture enables the model to learn semantic cues and contextual patterns that are often overlooked by frequency-based representations, making IndoBERT + CNN a powerful deep learning pipeline for understanding subtler differences between spam and non-spam messages.

By employing these two parallel feature extraction strategies, the study provides a meaningful comparison between traditional sparse representations (TF-IDF) and modern contextualized transformer-based embeddings (IndoBERT). This dual-pipeline design highlights the strengths and limitations of each approach in the context of Indonesian email spam detection, offering insights into how feature representation impacts model accuracy, generalization, and robustness across different types of spam content.

3.5. Modelling

Following feature extraction, the next phase of the project involves training two different classification models to evaluate their effectiveness in detecting spam emails: a TF-IDF–based CNN model and an IndoBERT–based CNN model. Before training, the dataset was divided into three subsets using an 80:10:10 split, consisting of training, validation, and testing sets. Stratified sampling was applied to maintain the original class distribution across all splits, ensuring a fair and consistent evaluation during model training and testing.

For the first pipeline, TF-IDF vectors were used as the numerical representation of text. After preprocessing and tokenization, the text corpus was transformed into TF-IDF feature matrices with a vocabulary size of 5,000. These vectors were reshaped into a 3-dimensional format and fed into a 1D Convolutional Neural Network (CNN). The CNN architecture consisted of a Conv1D layer for extracting n-gram features, followed by a max-pooling layer, dense layers, dropout, and finally a softmax classifier. The model was trained using the Adam optimizer and cross-entropy loss. This approach allows the CNN to learn local patterns from TF-IDF sequences, making it a strong baseline for comparison.

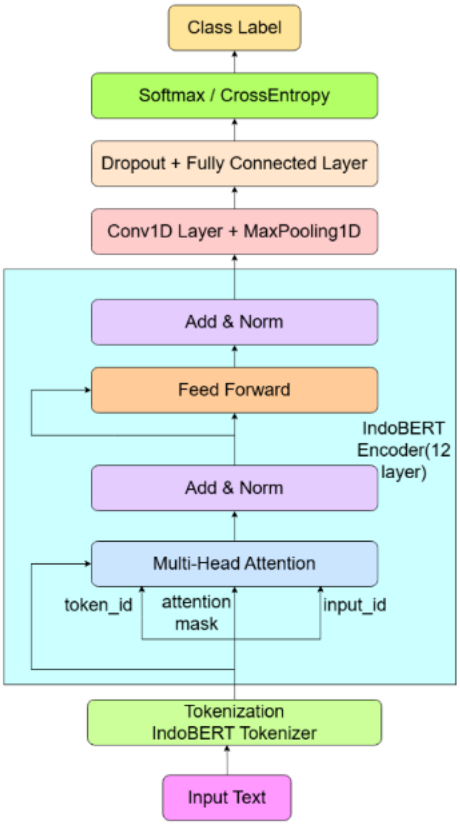

Fig. 2. IndoBERT Model Architecture

The second pipeline employs a more advanced deep learning model by integrating IndoBERT, a pretrained transformer model designed for the Indonesian language, with an additional CNN classifier. As illustrated in Figure 2, the input text is first processed using the IndoBERT tokenizer to generate input IDs and attention masks. These are passed into the IndoBERT encoder, producing contextual embeddings with rich semantic information. The output embeddings are then permuted and forwarded into a Conv1D layer to capture local sequence-level features. An adaptive max-pooling operation reduces the embedding dimension, followed by dropout and a fully connected layer that produces class logits. The final prediction is obtained using a softmax activation function.

During training, the IndoBERT encoder and CNN classifier were fine-tuned jointly using the Adam optimizer with a learning rate of 2e-5. All IndoBERT layers were kept trainable, enabling the model to adapt its contextual representations specifically for the spam classification task. This hybrid architecture effectively combines deep semantic understanding from IndoBERT with pattern detection capabilities of CNN, providing a strong comparison point against the TF-IDF-based CNN model.

3.6. Model Evaluation

In the final stage of the experiment, the performance of the trained models was evaluated using standard classification metrics commonly applied in text classification tasks, particularly spam detection. The evaluation process was conducted on the unseen test set, ensuring that the results reflect the true generalization capability of each model. Four primary metrics were used to assess performance: accuracy, precision, recall, and F1-score. These metrics collectively provide a comprehensive understanding of how well the model distinguishes between spam and non-spam emails.

Accuracy measures the overall proportion of correctly classified samples and offers a general performance overview. However, accuracy alone can be misleading when dealing with imbalanced datasets, where one class may dominate. Therefore, additional metrics were considered. Precision quantifies the proportion of emails predicted as spam that are truly spam, indicating the model's ability to avoid false positives. Recall, on the other hand, measures how many actual spam emails were successfully identified by the model, highlighting its ability to minimize false negatives. Finally, the F1-score, which is the harmonic mean of precision and recall, provides a balanced evaluation by considering both false positives and false negatives simultaneously.

For this project, all four metrics were computed using the macro-average approach to give equal importance to both spam and non-spam classes. This evaluation strategy ensures that the performance assessment remains fair and unbiased, especially in cases where class distribution is not perfectly balanced. The results from the TF-IDF + CNN model and the IndoBERT + CNN model were compared to determine the relative strengths and weaknesses of classical feature-based methods versus transformer-based architectures.

4. Evaluation

The performance comparison between the TF-IDF + CNN model and the IndoBERT + CNN model provides clear insights into how representation quality influences classification results. The TF-IDF + CNN model shows strong performance, demonstrating that even without contextual embeddings, convolutional architectures can effectively learn local n-gram patterns from sparse vector representations. This model achieves stable accuracy and gradually improving loss across training epochs, indicating consistent learning behavior. However, despite its effectiveness, TF-IDF remains limited by its inability to capture semantic meaning or contextual dependencies between words.

Table 1. Evaluation for the selected models.

Model	Accuracy	Precision	Recall	F1-Score
TF-IDF + CNN	0.5909	0.5971	0.5943	0.5890
IndoBERT + CNN	0.9810	0.9809	0.9811	0.9810

Building on these observations, the contrast between the two models becomes even more apparent when examining their training curves. The TF-IDF + CNN model exhibits moderate improvements across epochs, with training and validation accuracy fluctuating around the 0.60 range. This instability indicates that, while CNNs can extract local textual patterns from TF-IDF vectors, the sparse and context-agnostic nature of TF-IDF limits the model's ability to generalize. Similarly, the loss curves for TF-IDF + CNN decrease gradually but remain relatively high, suggesting slower convergence and a weaker ability to model deeper semantic relationships in the data.

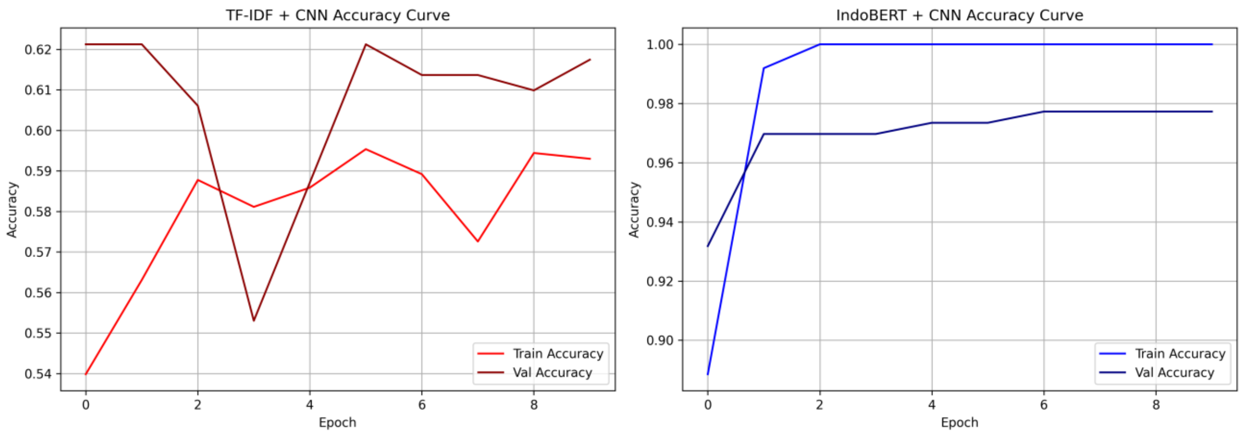

Fig. 3. Training Accuracy

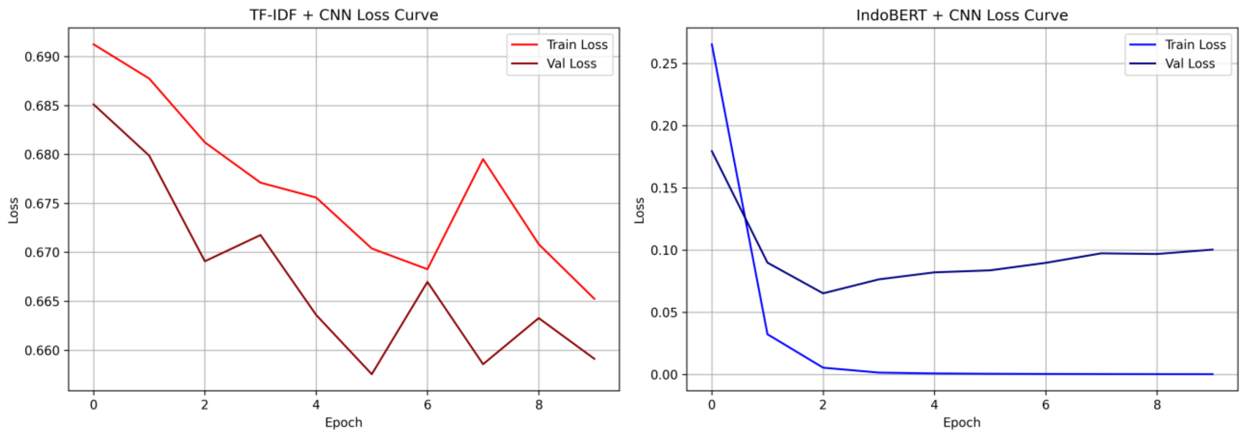

Fig. 4. Training Loss.

In comparison, the IndoBERT + CNN model shows a dramatically different learning behavior. Both training and validation accuracy rise sharply within the first few epochs, stabilizing at scores exceeding 0.97. The corresponding loss curve nearly reaches zero for the training set, while the validation loss remains consistently low. This rapid convergence highlights the strength of contextual embeddings produced by IndoBERT: because the representations already encode syntactic structure and semantic meaning, the CNN layers only need to refine local feature extraction rather than learn representations from scratch.

These differences underscore the crucial role of representation quality in modern NLP. TF-IDF relies solely on word frequency and cannot capture semantic similarity (e.g., “promo” and “diskon”), contextual meaning, or long-distance dependencies. As a result, models built on TF-IDF must compensate by learning more patterns directly from data, which is especially challenging for nuanced tasks like spam detection. IndoBERT, by contrast, leverages deep bidirectional transformers trained on large Indonesian corpora, enabling it to encode both sentence-level meaning and word-level relationships before any task-specific training begins.

Overall, the combined evidence from accuracy trends, loss behavior, and evaluation metrics clearly demonstrates that IndoBERT + CNN provides a more powerful and reliable approach for Indonesian spam classification. Its ability to integrate global semantic understanding with CNN-based local pattern extraction results in superior generalization and substantially higher performance compared to TF-IDF–based methods. This comparison illustrates how modern transformer-based language models have reshaped text classification pipelines, especially for languages with rich morphology and contextual variability such as Indonesian.

In contrast, the IndoBERT + CNN model significantly outperforms the TF-IDF approach due to its rich contextual embeddings generated by the IndoBERT encoder. Even early in training, the model converges rapidly, reaching high accuracy within the first few epochs. The combination of transformer-based embeddings and CNN's local feature extraction enables the model to capture both global semantic relationships and fine-grained textual patterns, resulting in superior generalization on validation data. Overall, IndoBERT + CNN demonstrates a clear advantage in terms of accuracy, precision, recall, and F1-score, highlighting the effectiveness of contextual language models when paired with deep learning classifiers.

5. Deployment

The deployment stage represents the final phase of this project, where the trained spam detection models are integrated into a functional application that can be accessed and used by end-users. The goal of deployment is to transform the machine learning models—previously developed and evaluated in a controlled environment—into a practical system capable of performing real-time predictions on new email inputs.

To achieve this, the TF-IDF + CNN and IndoBERT + CNN models were exported into serialized formats using joblib and TensorFlow's SavedModel API. These exported models were then loaded within a Streamlit-based web application, which serves as a lightweight, interactive interface for users. The deployed system allows users to input raw email text, select a preferred classification model, and receive instant predictions identifying whether the email is classified as SPAM or NON-SPAM.

During deployment, several preprocessing steps—such as cleaning, tokenization, and sequence transformation—were reproduced to ensure consistency with the training pipeline. For the TF-IDF model, the saved vectorizer was used to convert text into numerical features, while the IndoBERT model employed its pretrained tokenizer to encode input text into token IDs and attention masks. This guarantees that the deployed models interpret new inputs in the same format as the training data.

The Streamlit application was structured to be modular and scalable, allowing additional models to be integrated with minimal modification. Furthermore, deployment testing was conducted to verify model responsiveness, input compatibility, and inference stability. Although the current deployment setup runs locally, the system can be easily extended to cloud-based hosting environments such as Heroku, Render, or Vercel with minor adjustments, enabling broader accessibility and production-level usage.

Overall, the deployment process successfully bridges the gap between model development and user interaction, resulting in an accessible and functional spam detection platform that demonstrates the practical value of deep learning and NLP techniques in real-world applications.

6. Discussion

This study demonstrates a clear performance gap between classical feature-based deep learning approaches and transformer-based hybrid architectures for Indonesian email spam classification. The experimental results show that while the TF-IDF + CNN model is capable of learning discriminative patterns from textual data, its overall performance remains significantly lower than that of the IndoBERT + CNN model across all evaluation metrics.

The TF-IDF + CNN pipeline achieved moderate accuracy and F1-score, indicating that convolutional layers can still extract useful local n-gram patterns from sparse TF-IDF representations. This confirms prior findings that CNNs are effective at capturing local textual features even when contextual information is limited. However, the relatively low performance and unstable validation accuracy suggest that TF-IDF's bag-of-words nature restricts the model's ability to generalize, particularly when spam messages rely on paraphrasing, semantic similarity, or implicit intent rather than explicit keywords.

In contrast, the IndoBERT + CNN model achieved near-perfect performance, with accuracy, precision, recall, and F1-score all exceeding 0.98. This substantial improvement highlights the importance of contextualized language representations for spam detection tasks. IndoBERT embeddings encode semantic meaning, syntactic structure, and contextual dependencies, enabling the model to recognize spam patterns even when surface-level word usage varies. The CNN layers further enhance performance by extracting salient local features from the transformer outputs, resulting in a hybrid architecture that benefits from both global context and localized pattern detection.

The training behavior of the two models further supports these findings. The TF-IDF + CNN model exhibited slower convergence and higher loss values, indicating that it must learn most linguistic patterns directly from the data. Conversely, the IndoBERT + CNN model converged rapidly with consistently low loss, reflecting the advantage of pretrained representations that already capture rich linguistic knowledge before task-specific fine-tuning.

From a practical perspective, these results suggest a trade-off between computational efficiency and predictive performance. TF-IDF + CNN remains suitable for low-resource environments where computational constraints are strict and moderate accuracy is acceptable. However, for real-world Indonesian spam filtering systems where accuracy, robustness, and generalization are critical, transformer-based approaches such as IndoBERT + CNN are far more effective.

Overall, this discussion confirms that representation quality is a dominant factor in modern NLP pipelines. The superior performance of IndoBERT + CNN reinforces the shift toward contextual language models, particularly for low-resource languages like Indonesian, where semantic variability and informal language usage are common in spam content.

7. Limitation

Despite the strong performance achieved by the proposed models, this study has several limitations that should be acknowledged.

First, the dataset used in this research is based on a translated version of an English spam dataset. Although the translation preserves semantic intent, it may not fully capture the diversity, slang, and evolving linguistic patterns found in native Indonesian spam emails. As a result, the models' performance may be optimistic when evaluated on real-world, organically generated Indonesian spam data.

Second, the dataset size is relatively limited compared to large-scale industrial spam filtering systems. While sufficient for comparative analysis, a larger and more diverse dataset could further validate the robustness and generalization ability of the models, particularly the IndoBERT-based architecture.

Third, although IndoBERT + CNN achieved excellent results, the model is computationally expensive. Fine-tuning a transformer-based model requires significantly more memory and processing power compared to TF-IDF-based approaches. This limits its applicability in low-resource or real-time deployment scenarios without optimization techniques such as model pruning, quantization, or knowledge distillation.

Fourth, this study focuses primarily on standard evaluation metrics such as accuracy, precision, recall, and F1-score. Other important aspects, such as inference latency, memory consumption, robustness against adversarial spam, and long-term performance under concept drift, were not explored and remain important directions for future work.

Finally, the deployed application was tested in a local environment only. The system has not yet been evaluated under real-world operational conditions, such as high-volume email streams or continuously evolving spam patterns, which may affect long-term performance.

Addressing these limitations in future research could further strengthen the applicability and reliability of deep learning–based spam detection systems for Indonesian language contexts.

8. Reflection

8.1. Andrew

Through this project, I learned how significantly text representation choices influence the performance of deep learning models in Indonesian spam classification. Both TF-IDF + CNN and IndoBERT + CNN were able to learn meaningful patterns from the dataset. Implementing TF-IDF + CNN helped me understand how classical feature extraction methods can still serve as a solid and computationally efficient baseline. However, I also observed its limitations, particularly in capturing semantic and contextual information within Indonesian text.

Working with IndoBERT + CNN provided deeper insight into the importance of contextual embeddings. The model consistently achieved higher accuracy, precision, recall, and F1-score, which reinforced my understanding that transformer-based representations are more effective for language tasks involving nuanced meaning and context. Although I noticed slight overfitting during training, the IndoBERT-based model demonstrated strong generalization on unseen data. This experience taught me that higher model complexity does not necessarily reduce robustness when combined with appropriate training and evaluation strategies.

Beyond model development, this project also taught me the practical aspects of setting up a local deep learning environment. I learned how to properly configure a Conda environment, manage Python and library versions, and register kernels for use in tools such as VS Code and Jupyter Notebook. During development, I encountered several technical issues, including dependency conflicts, missing packages, and incorrect file or directory paths that caused the application or model to fail at runtime. Troubleshooting these problems required careful inspection of error messages, verification of directory structures, and consistent handling of relative and absolute paths when loading models, tokenizers, and vectorizers.

Through this debugging process, I developed a stronger understanding of how environment configuration and path management directly affect the reliability of a machine learning application. Resolving these issues improved my ability to diagnose errors systematically and ensured that the training pipeline, evaluation scripts, and deployed application were properly connected to the required resources.

Overall, this project helped me reflect on the trade-offs between model complexity, computational cost, and performance, while also strengthening my practical skills in environment setup and troubleshooting. While TF-IDF + CNN remains useful as a lightweight baseline, IndoBERT + CNN is more suitable for real-world Indonesian spam detection scenarios where accuracy and contextual understanding are crucial. This project strengthened my appreciation for selecting the right feature representation and building a stable development environment based on the problem domain and deployment constraints.

8.2. Nicholas Wira Angkasa

Working on this project has been a valuable learning experience that strengthened both my technical and analytical skills. Throughout the development of the TF-IDF + CNN and IndoBERT + CNN models, I gained a deeper understanding of how different text-representation techniques influence model performance, training behaviour, and generalization.

One key lesson I learned is that model complexity does not always guarantee better results. TF-IDF, despite being a traditional method, performed quite well when combined with CNN, proving that classical techniques can still be highly effective when applied correctly. At the same time, implementing IndoBERT taught me the importance of leveraging pre-trained language models and understanding how they capture semantic relationships far beyond surface-level patterns.

During the project, I also improved my ability to troubleshoot issues, such as data-cleaning inconsistencies, incorrect vector shapes, model-loading problems, and Git version control conflicts. These challenges helped me become more disciplined in structuring code, documenting experiments, and organizing files to ensure reproducibility.

Another important reflection is the value of experiment tracking and visualization. By plotting confusion matrices, training curves, and comparing metrics side-by-side, I realized how crucial it is to evaluate models from multiple perspectives instead of relying solely on accuracy.

Finally, this project helped me grow not only as a developer but also as a learner. It taught me patience, problem-solving, and the importance of continuously iterating until the system works as expected. I now feel more confident in handling NLP workflows, deep learning pipelines, and experiment design, which I believe will be highly beneficial for my future academic and professional journey.

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